Section Report 2 — WGS: Reads to Info

BIOL 4810

2026-03-12

What you are doing: You are given **raw reads only**. You must run the full WGS workflow and document it in a way that is reproducible and ready to commit to Git.
Submission: Fill this page out, upload the requested screenshots/PDFs into the page, then use **Download as PDF (Print)**.
Notes: You should taking detailed notes about each step, what software you are using, what version of the software (if available), and output files so that someone else could theoretically repeat your exact work.

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0) Required folder structure (make this first)

Inside your course repository, create a folder named something like:

```
sectionreport2_wgs/
```

Inside that folder, create:

```
rawfastqc/
fastp/
trimmedfastqc/
assembly/
quast/
filteredassembly/
checkm2/
prokka/
ani/
tygs/
miningfiles/
```

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1) FASTQC on raw reads (Galaxy)

In Galaxy 1. Upload your raw reads (make sure to create the paired data set) 2. Run **FastQC** on **raw** reads 3. Open the FastQC HTML report

Download requirement: Download the FastQC HTML report for **either R1 or R2** and save it in: `rawfastqc/`

1A. FastQC version (from Galaxy Job Info):

FastQC v0.74+galaxy1 (Galaxy wrapper)

1B. Read length reported:

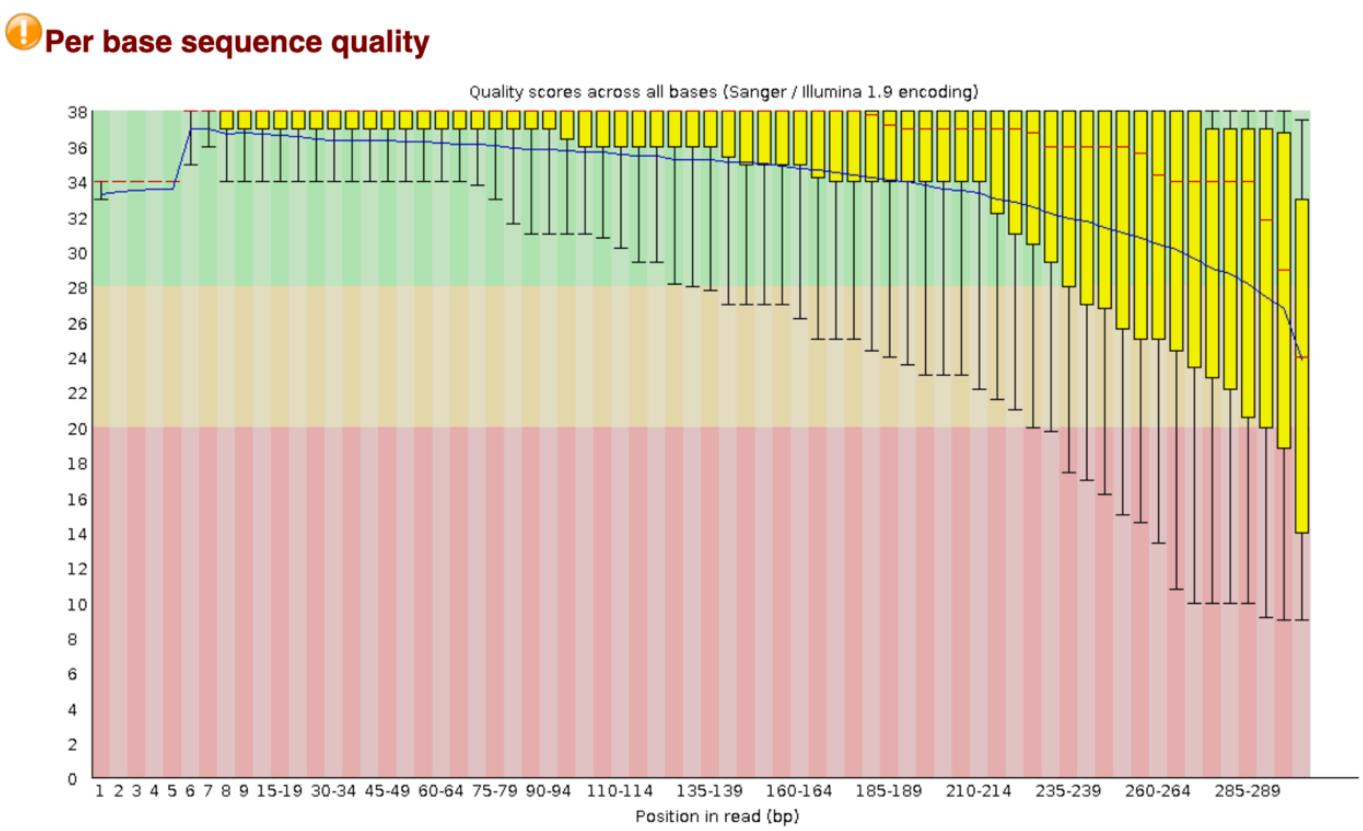
35-301 bp

1C. One key quality issue you see in raw reads (2–3 sentences):

Further along in the read, quality scores seem to dip and vary with a wider range than throughout lower positions in the read. Base calling can become less accurate toward the end of a read due to the presence of low quality bases.

Upload screenshot: raw FastQC quality plot (R1 or R2)

Choose File rawFastQC



Save the downloaded FastQC HTML report in `rawfastqc/` using your naming convention.

2) fastp trimming/filtering (Galaxy)

In Galaxy 1. Run **fastp** on your raw reads 2. Keep defaults unless your instructor specifies otherwise 3. Make sure the output includes the report

Download requirement: Download the **fastp report HTML** and save it in: `fastp/`

2A. fastp version (from Galaxy Job Info):

fastp v1.1.0+galaxy0 (Galaxy wrapper)

2B. What did fastp change? (2–3 sentences)

Fastp removed reads with low quality and too many N.

2C. Name one metric that changed from BEFORE → AFTER in the fastp report (e.g., Q30%, GC%, read length distribution) and state how it changed:

After filtering with fastp, Q30 decreased from 444.4 M bases to 425.9 M. Total bases decreased after filtering from about 597 M to 543 M bases.

Upload your fastp HTML report (downloaded from Galaxy)

Choose File fastp.html

Selected file: fastp.html

Save your fastp HTML in `fastp/` using your naming convention.

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3) FASTQC on trimmed reads (Galaxy)

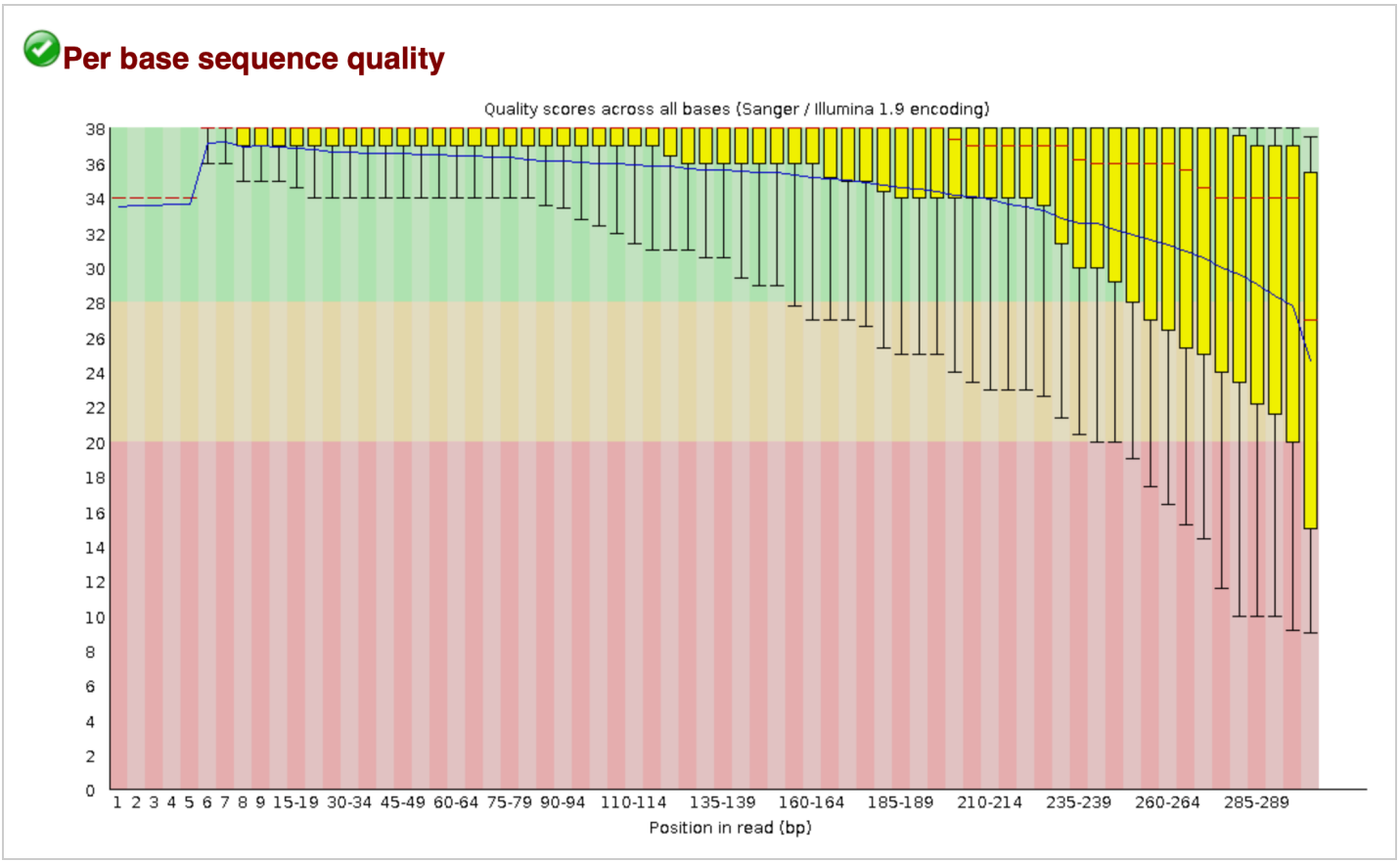
In Galaxy 1. Run **FastQC** on the **trimmed** reads 2. Download the FastQC HTML report for either R1 or R2 and save it in: `trimmedfastqc/`

3A. One key improvement (or remaining issue) after trimming (2–3 sentences):

Descending base call accuracy toward the end of the read remained an issue after trimming. The graph of quality scores mirrors that of the raw read.

Upload screenshot: trimmed FastQC quality plot (R1 or R2)

Choose File trimmedFa…orwardR1



Save the downloaded FastQC HTML report in `trimmedfastqc/` using your naming convention.

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4) Assembly (SPAdes in Galaxy)

In Galaxy 1. Run **SPAdes** on your trimmed reads 2. Download the **contigs FASTA** and save it in: `assembly/`

4A. SPAdes version (from Galaxy Job Info):

SPAdes v4.2.0+galaxy0 (Galaxy wrapper)

4B. Name of your downloaded contigs file (after you renamed it):

FilterFASTAcontigs.fasta

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5) Assembly evaluation (QUAST in Galaxy)

In Galaxy 1. Run **QUAST** on your SPAdes contigs 2. Download the **QUAST PDF report only** and save it in: `quast/`

5A. QUAST version (from Galaxy Job Info):

QUAST v5.3.0+galaxy1 (Galaxy wrapper)

5B. N50:

305484

5C. L50:

6

5D. Largest contig:

753920

5E. Total assembly length:

5380373

Upload your QUAST PDF report (downloaded from Galaxy)

Choose File Quast.pdf

Selected file: Quast.pdf

Save the QUAST PDF in `quast/` using your naming convention.

5F. Based on QUAST, is your assembly fragmented? Use at least 2 metrics to justify (2–4 sentences):

The relatively high N50 value indicates that the assembly is more contiguous, as 50% of the contigs are this long or longer in bp. The low L50 value also indicates that the assembly is more contiguous as only 6 contigs contain 50% of the genome.

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6) Genome quality (CheckM2 in Galaxy)

6A) Filter FASTA (Galaxy)

To reduce tiny contigs, create a filtered assembly for work going forward:

In Galaxy 1. Use **Filter FASTA** 2. Input: your SPAdes contigs FASTA 3. Filter by minimum contig length: **500 bp** 4. Save the filtered contigs as a new dataset in `filteredcontigs`

6A. Confirm your filter rule (one line):

Kept contigs at or above 500 bp

6B) In Galaxy

1. Run **CheckM2** on your assembly (contigs)
2. Download the CheckM2 report/output and save it in: `checkm2/`

6B. CheckM2 version (from Galaxy Job Info):

CheckM2 v1.1.0+galaxy0 (Galaxy wrapper)

6C. Completeness (%):

98.98%

6D. Contamination (%):

0.08%

6E. Interpret these numbers: is this "good enough" for downstream annotation? Why? (2–3 sentences)

would trust this assembly for annotation and species identification. These scores suggest that the assembly likely represents one species, whose genome was mostly recovered and
thus qualified for downstream analysis.

Folder requirement: Put the downloaded CheckM2 report/output in `checkm2/` with your naming convention.

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7) Prokka annotation (Galaxy)

In Galaxy 1. Run **Prokka** 2. Input: your **filtered** contigs (≥500 bp) 3. Change at least one setting from default as instructed (record it below) 4. Run Prokka

7A. Prokka version (from Galaxy Job Info):

Galaxy version 1.14.6+galaxy1

7B. What did you change from Prokka defaults? (1–2 sentences)

Force GenBank Compliance-- Yes

Download requirements (Prokka): download and save these into `prokka/` - `.txt` (summary) - `.gbk` (GenBank format annotation) - `.ffn` (nucleotide sequences of features)

7C. Paste your Prokka summary lines (from the .txt) and describe why this does or does not look like a

typical bacterial genome:

bases: 5380373
CDS: 5030
gene: 5119
rRNA: 10
tRNA: 78
tmRNA: 1

7D. How many hypothetical proteins does your organism have? What percent of CDS are hypothetical?

% hypothetical proteins of CDS= hypothetical proteins/CDS= 1196/5030=0.238 (23.8%)

rRNA feature check (from Prokka outputs)

Open your Prokka outputs and locate the rRNA entries (typically in the `.ffn` and/or `.tbl` / `.gff` depending on Galaxy wrapper).

7E. Paste the description/header lines for 23S, 16S, and 5S rRNA (three lines total):

>NMDDALNH_05098 23S ribosomal RNA (partial)
>NMDDALNH_05102 16S ribosomal RNA (partial)
>NMDDALNH_04545 5S ribosomal RNA (partial)

7F. Paste the full 16S rRNA FASTA sequence (header + sequence):

TTGAAGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTC
GAGCGGTAGCACAGAGAGCTTGCTCTCGGGTGACGAGCGGCGGACGGGTGAGTAATGTCT
GGGAAACTGCCTGATGGAGGGGGATAACTACTGGAAACGGTAGCTAATACCGCATAACGT
CGCAAGACCAAAGTGGGGGACCTTCGGGCCTCATGCCATCAGATGTGCCCAGATGGGATT
AGCTAGTAGGTGGGGTAACGGCTCACCTAGGCGACGATCCCTAGCTGGTCTGAGAGGATG
ACCAGCCACACTGGAACTGAGACACGGTCCAGACTCCTACGGGAGGCAGCAGTGGGGAAT
ATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCCGCGTGTGTGAAGAAGGCCTTCGGG
TTGTAAAGCACTTTCAGCGGGGAGGAAGGCGTTAAGGTTAATAACCTTGGCGATTGACGT
TACCCGCAGAAGAAGCACCGGCTAACTCCGTGCCAGCAGCCGCGGTAATACGGAGGGTGC
AAGCGTTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTCTGTCAAGTCGGATGT

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8) 16S species ID using BLAST (NCBI)

Use your 16S rRNA FASTA from above.

8A. Top hit species name + percent identity + query coverage:

A

8B. Based on 16S alone, how confident are you in species-level ID? Why? (2–3 sentences)

For the top hit, we can infer genus level similarity because both identity and query coverage are above 97-99%. Anything over 99% can be interpreted as the two species in comparison likely being the same.

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9) ANI (EZBioCloud) — two comparisons

Use the ANI tool and compare your assembly to **two reference genomes** that you choose (based on your best guess from 16S/Prokka/TYGS). Open the site here (new tab): EZBioCloud ANI (https://www.ezbiocloud.net/tools/ani)

Download requirement: download the two reference genomes you used and save them in `ani/` with clear names.

9A. Reference genome #1 (species/strain):

Klebsiella pneumoniae strain DSM 30104

9B. ANI % vs reference #1:

99.14%

9C. Reference genome #2 (species/strain):

Klebsiella variicola strain F2R9

9D. ANI % vs reference #2:

94.68%

9E. Interpret your ANI: do you meet a typical same-species ANI threshold? What conclusion do you draw? (2–4 sentences)

Following the rule of thumb that ANI values greater than or equal to 95-96% are presumably the same species, and those under 95% are different species, I conclude that the first reference genome #1 is the same as my species. Secondly, reference genome #2 is a different species than mine because it falls short of 95%.

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10) TYGS phylogeny (use filtered contigs for this report)

Run TYGS using your filtered contigs (≥500 bp). Export/download the tree (.phy file) and view in iTOL. Make sure that you have displayed the bootstrap values. Export your tree as a .pdf file. The .phy and .pdf should be in your `tygs/` folder

In your own words: what kinds of evidence does TYGS use to place your genome? Did you need to generate a proteome tree or was the genome tree sufficient? Did it correspond to what you found with your 16S BLAST results.

Unlike BLAST, which tends to be ran against the NCBI 16S database and is therefore limited to a single 16S gene for interpretation, TYGS compares entire genomes to validated strains. [Answers to remaining questions NA: TYGS down].

Upload your TYGS/iTOL tree as a PDF

Choose File tygs_error

TYPE (STRAIN) GENOME SERVER | Submit your query | News | Examples | Background | DSMZ Digital Diversity | DSMZ

User request

Thank you for your submission! A confirmation e-mail will arrive soon (hint: check your spam folder, just in case)!

Job ID	Submission date	Contact e-mail	Num. query genomes	Restrict to query genomes	Num. reference genomes	Result links
1d7576cd-759a-4a79-afc5-bf8d29bd0d93	2026-03-24 18:13:55 +0100	anames@csustan.edu	1	no	0	Results

#	Genome (user-provided or manually selected on submission)	Source
1	FilterFASTAcontigs.fasta	per file

11) Genome mining (Day 14 web tools) — ALL REQUIRED

You must run **all** tools below and include a **screenshot of results** for each. Save screenshots in: `miningfiles/` with clear names.

Also: record the **version/date/database** if the tool reports it (many do on the results page or in the footer).

11A) CARD / RGI (AMR genes)

Open: CARD / RGI (https://card.mcmaster.ca/analyze/rgi)

11A1. CARD/RGI version + database/date info (copy from results page if shown):

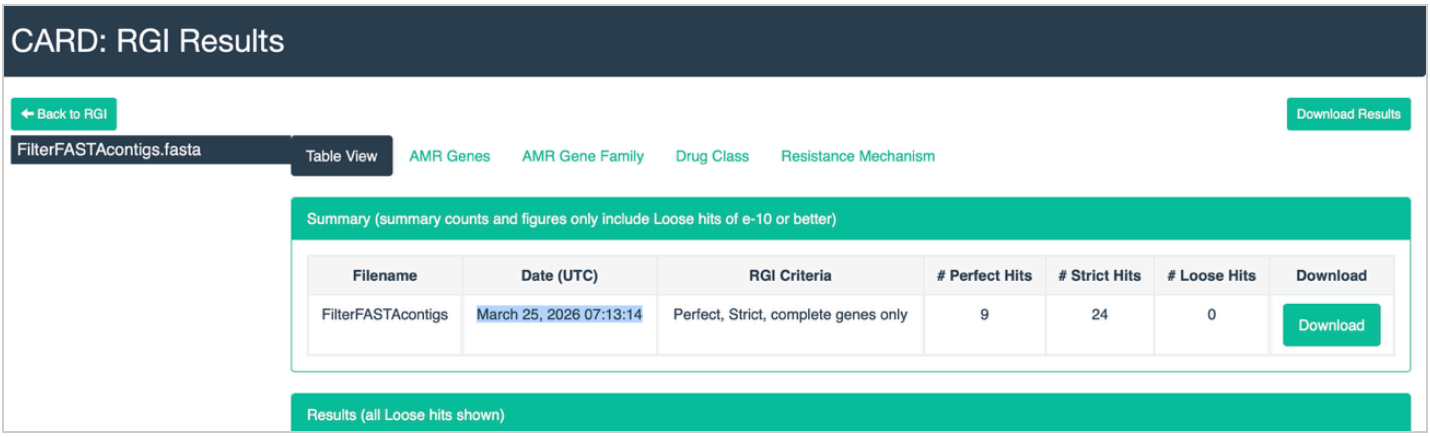
RGI v6.0.5, CARD v4.0.1; March 25, 2026 07:13:14

11A2. Key AMR findings (2–4 sentences). Include at least one gene + evidence type (e.g., perfect/strict/loose):

AMR findings include a perfect hit for gene oqxA, which provides resistance to antibiotics including fluoroquinolone, glycylcylcine, tetracycline, diaminopyrimidine, and nitrofuran via antibiotic efflux. Additionally, there was a perfect hit for gene SHV-27 which confers resistance to cephalosporins and penicillin beta-lactam via antibiotic activation.

Upload screenshot: CARD/RGI results

Choose File Klebsiellap…ardrgi.png



Save as something like `SampleID_card_rgi.png` in `miningfiles/` .

11B) ResFinder (AMR genes)

Open: ResFinder (https://genepi.food.dtu.dk/resfinder)

11B1. ResFinder version/database info (copy from results page if shown):

ResFinder-2.5.1

11B2. Key findings (2–4 sentences). Do ResFinder and CARD agree? If not, why might they differ?

ResFinder and CARD do agree! CARD garnered 9 perfect hits and ResFinder, which highlights acquisition through HGT over mutation, also garnered 9 AMR gene hits. CARD outputs are more exhaustive than ResFinder and included strict (less than perfect) AMR hits as well.

Upload screenshot: ResFinder results

Choose File Klebsiellap…finder.png

Acquired AMR gene hits Hide

Resistance gene	Identity	Contig	Phenotype(s)	PMID	Accession no.
blaSHV-27	100.00 %	NODE_9_length_223764_cov_42.108886	Amoxicillin, Ampicillin, Cefepime, Cefotaxime, Ceftazidime, Piperacillin, Aztreonam, Ticarcillin, Ceftriaxone	11266422	AF293345
blaLAP-2	100.00 %	NODE_22_length_33968_cov_85.612009	Amoxicillin, Ampicillin, Piperacillin, Ticarcillin, Cephalothin	18550213	EU159120
fosA6	99.29 %	NODE_14_length_148325_cov_47.624502	Fosfomycin	unpublished	KU254579
OqxB	99.05 %	NODE_1_length_753920_cov_47.517281	Ciprofloxacin, Nalidixic acid, Trimethoprim, Chloramphenicol	18440636	EU370913
OqxA	99.15 %	NODE_1_length_753920_cov_47.517281	Ciprofloxacin, Nalidixic acid, Trimethoprim, Chloramphenicol	18440636	EU370913
qnrS1	100.00 %	NODE_22_length_33968_cov_85.612009	Ciprofloxacin	18426894	AB187515
sul1	100.00 %	NODE_22_length_33968_cov_85.612009	Sulfamethoxazole	unpublished	U12338
tet(A)	100.00 %	NODE_22_length_33968_cov_85.612009	Tetracycline, Doxycycline	12654659	AJ517790
dfrA1	100.00 %	NODE_22_length_33968_cov_85.612009	Trimethoprim	6308574	X00926

Input parameters

Input files	Acquired antimicrobial resistance genes	Species and input data type	Database versions
Input fasta: FilterFASTAcontigs.fasta	Threshold for ID: 80% Minimum length: 60%	Selected species: other Input type: FASTA	ResFinder-2.5.1

Save as `SampleID_resfinder.png` in `miningfiles/` .

11C) PathogenFinder 2.0 (pathogenicity prediction)

Open: PathogenFinder (https://genepi.food.dtu.dk/pathogenfinder)

11C1. PathogenFinder version/database info (copy from results page if shown):

PathogenFinder2-0.6.0

11C2. Report the predicted probability/label and interpret it (2–4 sentences):

Values exceeding 0.5 indicate pathogenic capacity. The mean predicted probability for pathogenic capacity here is 0.97, which can interpreted as pathogenic capacity.

Upload screenshot: PathogenFinder results

Choose File Klebsiellap…finder.png

Results

Download

Software version: PathogenFinder2-0.6.0

Session ID: zklmap5td4cccka2hbwi3tjpqh7525p1

Bacterial Pathogenic Capacity prediction

The pathogenic capacity of each bacterial genome is estimated using an ensemble of four neural networks, each trained on a different subset of the original dataset. The final prediction is computed as the mean output of these models. Values above 0.5 indicate pathogenic capacity, whereas values below this threshold indicate the absence of such capacity. Scores close to 0.5 reflect samples that lie near the model's decision boundary, meaning the model identifies mixed or ambiguous features that do not strongly support either class.

Module	Prediction
Neural Network 1	0.9819
Neural Network 2	0.9399
Neural Network 3	0.9761
Neural Network 4	0.9888
Mean	**0.9717 (0.0189)**

The bacterial genome is predicted to have *Human Pathogenic* capacity

Save as `SampleID_pathogenfinder.png` in `miningfiles/`.

11D) VFDB virulence factors via **ABRicate** (Galaxy)

The VirulenceFinder/VFDB web outputs can be inconsistent, so for this report you will run VFDB searches **inside Galaxy** using **ABRicate**.

In Galaxy (ABRicate): 1. Search tools for **ABRicate** 2. **Input:** your **Prokka GenBank file** (`.gbk`) 3. Under**Advanced**, **Database:** choose **VFDB** 4. Run ABRicate 5. Download the **tabular** results file (`.tabular`) and save it in `miningfiles/` using your naming convention.

11D1. ABRicate version + VFDB database version (from Galaxy Job Info):

ABRicate Galaxy Version 1.0.1

11D2. How many VFDB hits did you get (from the results table/summary)?

10

11D3. List your top 3 VFDB hits (gene name + brief description/function). If you have fewer than three, list all. If you have none, you can put NA:

Accession NP_286011; (ykgK/ecpR) regulator protein EcpR [ECP (VF0404)] [Escherichia coli O157:H7 str. EDL933]
Accession NP_286010; (yagZ/ecpA) E. coli common pilus structural subunit EcpA [ECP (VF0404)] [Escherichia coli O157:H7 str. EDL933]

Upload your ABRicate VFDB results table (tabular)

Choose File Klebsiellap…db.tabular

Selected file: Klebsiellapneumoniae_abricate_vfdb.tabular

Save as `SampleID_abricate_vfdb.tabular` in `miningfiles/` .

11E) antiSMASH (biosynthetic gene clusters)

Open: antiSMASH (https://antismash.secondarymetabolites.org/)

11E1. antiSMASH version/database info (copy from results page if shown):

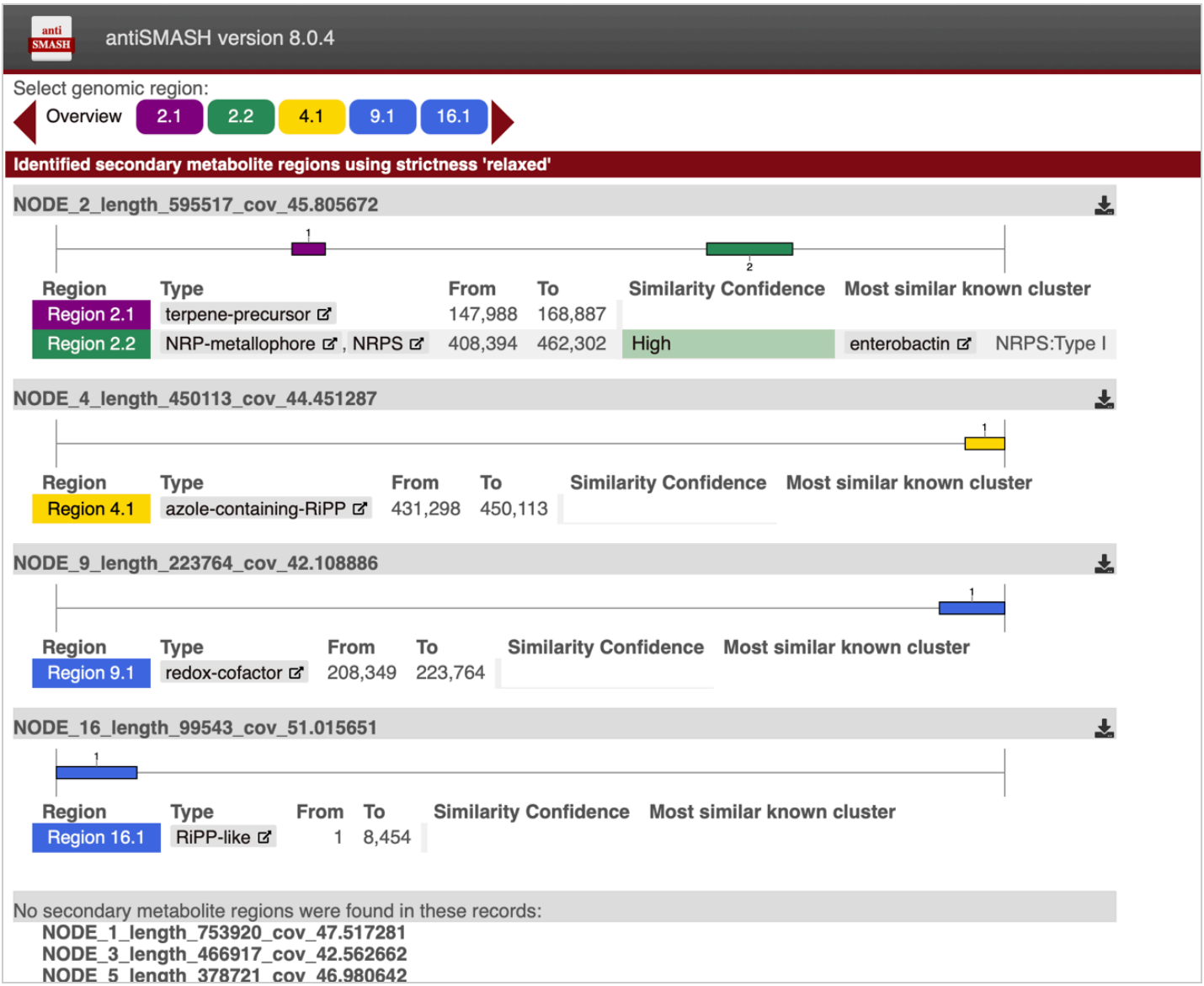
antiSMASH v8.0.4

11E2. Key BGC finding(s) (2–4 sentences). Include at least one cluster type and whether it matches a known cluster:

Five BCGs were found for secondary metabolites including terpene-precursor, NRP-metallophore and NRPS, azalea-containing RiPP, redox-cofactor, and RiPP-like. NRP-metallophore and NRPS presented a high similarity confidence to known enterobactin cluster.

Upload screenshot: antiSMASH results

Choose File Klebsiella…ismash.png



Save as `SampleID_antismash.png` in `miningfiles/` .

11F) IslandViewer (genomic islands)

Open: IslandViewer (https://www.pathogenomics.sfu.ca/islandviewer/)

11F1. IslandViewer version/database info (copy from results page if shown):

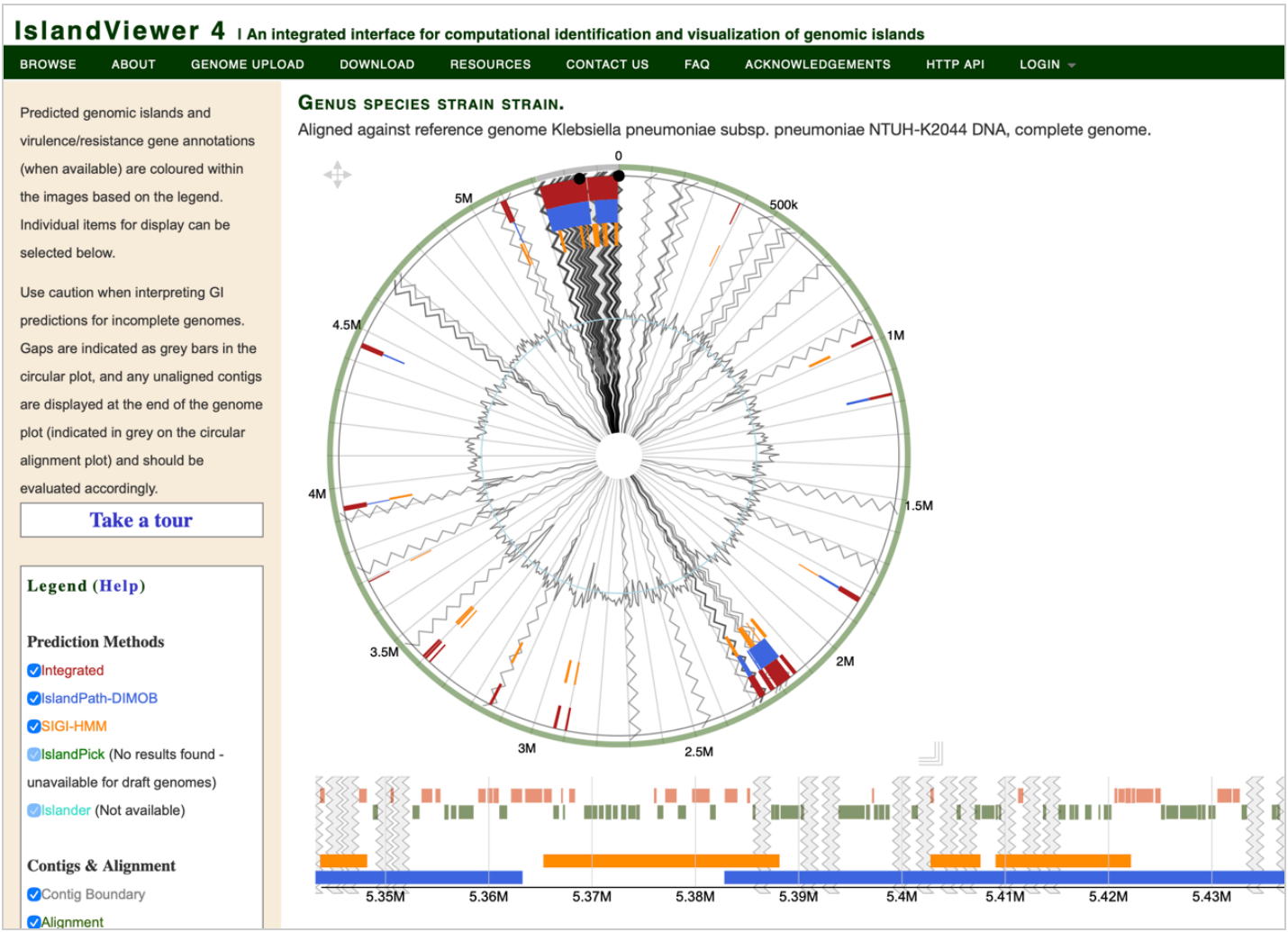
IslandViewer 4

11F2. Key island finding(s) (2–4 sentences). How many predicted islands? Any notable features (mobility genes, tRNA insertion sites, etc.)?

Roughly 60 islands were predicted by IslandViewer. Notable features include heat shock proteins, multidrug resistance proteins, and a gene for arsenate reductase.

Upload screenshot: IslandViewer results

Choose File Klebsiella…viewer.png



Save as `SampleID_islandviewer.png` in `miningfiles/`.

11G) CRISPRCasFinder (CRISPR arrays + Cas genes)

Open: CRISPRCasFinder (https://crisprcas.i2bc.paris-saclay.fr/)

11G1. CRISPRCasFinder version/database info (copy from results page if shown):

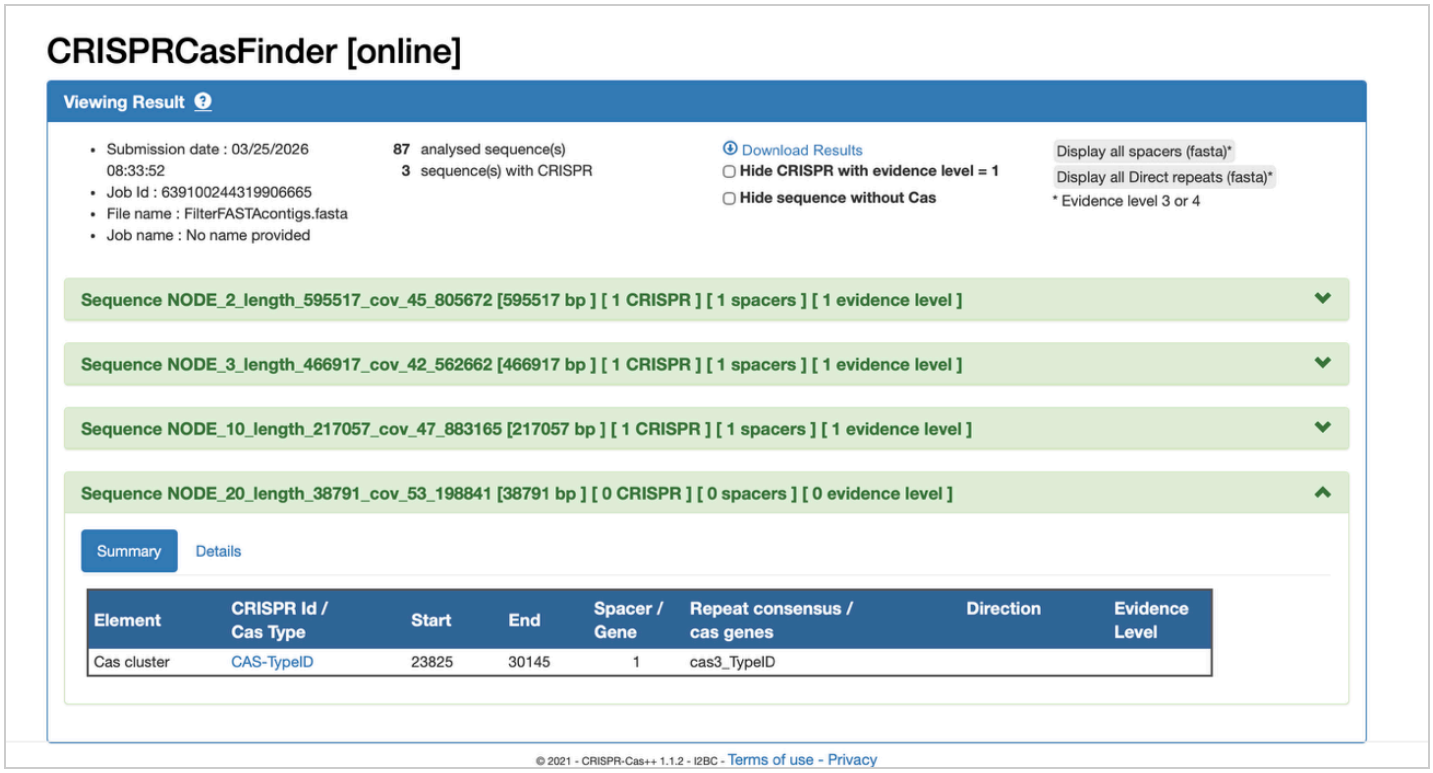
CRISPRCasFinder v1.1.2

11G2. Key CRISPR/Cas findings (2–4 sentences). Did you find CRISPR arrays? Cas genes? What confidence level?

Three CRISPR arrays were found, all of which had an evidence level of 1. A single Cas gene was found, which was not associated with an evidence level.

Upload screenshot: CRISPRCasFinder results

Choose File Klebsiellap…finder.png



Save as `SampleID_crisprcasfinder.png` in `miningfiles/` .

11H) Synthesis across genome mining tools

11H1. In 6–10 sentences, synthesize what genome mining suggests about this organism (resistance potential, virulence potential, mobile elements, secondary metabolism, etc.). Point to at least one result from each tool:

related genes. One of the four secondary metabolite regions identified by antiSMASH presented with a high similarity confidence to a known cluster– it is unclear whether identified BGCs reflect pathogenicity of the organism and its genome. While not proof of pathogenicity alone, IslandViewer yielded notable features including heat shock proteins, multidrug resistance proteins, and a gene for arsenate reductase which at least together indicate resistance to stress and persistence in unfavorable environments. The CRISPRCasFinder results indicate that the organism belonging to this genome encountered and adapted to phages in its history. While not a direct virulence factor, CRISPRCas systems are linked to the expression of virulence factors used for integrity and survival.

12) Final synthesis (your conclusion)

12A. What is your best-supported species ID? Use at least 3 evidence types (16S, ANI, TYGS, CheckM2, Prokka) and explain remaining uncertainty (8–12 sentences):

Relatively high N50 and low L50 values from QUAST indicate that the assembly is more contiguous. CheckM2 completeness and contamination over 98% and under 1%, respectively, support trustworthiness of the filtered assembly for subsequent annotation and species identification, as the scores suggest that the assembly likely represents one species whose genome was mostly recovered. Prokka estimated that 23.8% of CDS were hypothetical proteins, which was low enough to indicate higher confidence in the annotation for downstream analysis. NCBI Blastn 16S species identification metrics delivered confidence for a species-level ID. Top hit Klebsiella pneumoniae strain DSM 30104 had a 99.70% identity and 99% query coverage. The use of an alternate, more distantly related reference genome in ANI affirmed 16S results in that the unknown organism and Klebsiella pneumoniae reference genome garnered an OrthoANI value of 99.14%

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